

LeetCode Array Practice in C

15 beginner problems • Simple C Solutions

Important: These are normal C programs using main(). No malloc, no pointer return values, and no LeetCode function signatures.

1. LeetCode 1480 — Running Sum of 1d Array

Simple idea: Solve the problem using basic arrays, loops and conditions.

```
#include <stdio.h>

int main() {
    int n, arr[100];

    scanf("%d", &n);
    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    for (int i = 1; i < n; i++)
        arr[i] = arr[i] + arr[i - 1];

    for (int i = 0; i < n; i++)
        printf("%d ", arr[i]);

    return 0;
}
```

Student focus: Understand the loop and array indexing first. Optimization can be taught later.

2. LeetCode 1470 — Shuffle the Array

Simple idea: Solve the problem using basic arrays, loops and conditions.

```
#include <stdio.h>

int main() {
    int n, arr[100], ans[100];

    scanf("%d", &n);
    for (int i = 0; i < 2 * n; i++)
        scanf("%d", &arr[i]);

    int k = 0;
    for (int i = 0; i < n; i++) {
        ans[k++] = arr[i];
        ans[k++] = arr[i + n];
    }

    for (int i = 0; i < 2 * n; i++)
        printf("%d ", ans[i]);

    return 0;
}
```

Student focus: Understand the loop and array indexing first. Optimization can be taught later.

3. LeetCode 1431 — Kids With the Greatest Number of Candies

Simple idea: Solve the problem using basic arrays, loops and conditions.

```
#include <stdio.h>

int main() {
    int n, arr[100], extra;
    scanf("%d", &n);

    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    scanf("%d", &extra);

    int max = arr[0];
    for (int i = 1; i < n; i++)
        if (arr[i] > max) max = arr[i];

    for (int i = 0; i < n; i++)
        printf("%s ", arr[i] + extra >= max ? "true" : "false");

    return 0;
}
```

Student focus: Understand the loop and array indexing first. Optimization can be taught later.

4. LeetCode 1672 — Richest Customer Wealth

Simple idea: Solve the problem using basic arrays, loops and conditions.

```
#include <stdio.h>

int main() {
    int rows, cols, a[20][20];

    scanf("%d %d", &rows, &cols);

    for (int i = 0; i < rows; i++)
        for (int j = 0; j < cols; j++)
            scanf("%d", &a[i][j]);

    int max = 0;

    for (int i = 0; i < rows; i++) {
        int sum = 0;
        for (int j = 0; j < cols; j++)
            sum += a[i][j];

        if (sum > max) max = sum;
    }

    printf("%d", max);
    return 0;
}
```

Student focus: Understand the loop and array indexing first. Optimization can be taught later.

5. LeetCode 1920 — Build Array from Permutation

Simple idea: Solve the problem using basic arrays, loops and conditions.

```
#include <stdio.h>

int main() {
    int n, arr[100], ans[100];

    scanf("%d", &n);
    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    for (int i = 0; i < n; i++)
        ans[i] = arr[arr[i]];

    for (int i = 0; i < n; i++)
        printf("%d ", ans[i]);

    return 0;
}
```

Student focus: Understand the loop and array indexing first. Optimization can be taught later.

6. LeetCode 1365 — How Many Numbers Are Smaller Than the Current Number

Simple idea: Solve the problem using basic arrays, loops and conditions.

```
#include <stdio.h>

int main() {
    int n, arr[100], ans[100];

    scanf("%d", &n);
    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    for (int i = 0; i < n; i++) {
        ans[i] = 0;
        for (int j = 0; j < n; j++)
            if (arr[j] < arr[i])
                ans[i]++;
    }

    for (int i = 0; i < n; i++)
        printf("%d ", ans[i]);

    return 0;
}
```

Student focus: Understand the loop and array indexing first. Optimization can be taught later.

7. LeetCode 1295 — Find Numbers with Even Number of Digits

Simple idea: Solve the problem using basic arrays, loops and conditions.

```
#include <stdio.h>

int main() {
    int n, arr[100];
    scanf("%d", &n);

    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    int count = 0;

    for (int i = 0; i < n; i++) {
        int x = arr[i], digits = 0;

        while (x > 0) {
            digits++;
            x /= 10;
        }

        if (digits % 2 == 0)
            count++;
    }

    printf("%d", count);
    return 0;
}
```

Student focus: Understand the loop and array indexing first. Optimization can be taught later.

8. LeetCode 1512 — Number of Good Pairs

Simple idea: Solve the problem using basic arrays, loops and conditions.

```
#include <stdio.h>
```

```
int main() {  
    int n, arr[100];  
    scanf("%d", &n);  
  
    for (int i = 0; i < n; i++)  
        scanf("%d", &arr[i]);  
  
    int count = 0;  
  
    for (int i = 0; i < n; i++)  
        for (int j = i + 1; j < n; j++)  
            if (arr[i] == arr[j])  
                count++;  
  
    printf("%d", count);  
    return 0;  
}
```

Student focus: Understand the loop and array indexing first. Optimization can be taught later.

9. LeetCode 2006 — Count Number of Pairs With Absolute Difference K

Simple idea: Solve the problem using basic arrays, loops and conditions.

```
#include <stdio.h>

int main() {
    int n, arr[100], k;
    scanf("%d", &n);

    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    scanf("%d", &k);

    int count = 0;

    for (int i = 0; i < n; i++) {
        for (int j = i + 1; j < n; j++) {
            int diff = arr[i] - arr[j];
            if (diff < 0) diff = -diff;

            if (diff == k)
                count++;
        }
    }

    printf("%d", count);
    return 0;
}
```

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10. LeetCode 1 — Two Sum

Simple idea: Solve the problem using basic arrays, loops and conditions.

```
#include <stdio.h>

int main() {
    int n, arr[100], target;
    scanf("%d", &n);

    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    scanf("%d", &target);

    for (int i = 0; i < n; i++) {
        for (int j = i + 1; j < n; j++) {
            if (arr[i] + arr[j] == target) {
                printf("%d %d", i, j);
                return 0;
            }
        }
    }

    return 0;
}
```

Student focus: Understand the loop and array indexing first. Optimization can be taught later.

11. LeetCode 217 — Contains Duplicate

Simple idea: Solve the problem using basic arrays, loops and conditions.

```
#include <stdio.h>

int main() {
    int n, arr[100];
    scanf("%d", &n);

    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    for (int i = 0; i < n; i++) {
        for (int j = i + 1; j < n; j++) {
            if (arr[i] == arr[j]) {
                printf("true");
                return 0;
            }
        }
    }

    printf("false");
    return 0;
}
```

Student focus: Understand the loop and array indexing first. Optimization can be taught later.

12. LeetCode 27 — Remove Element

Simple idea: Solve the problem using basic arrays, loops and conditions.

```
#include <stdio.h>

int main() {
    int n, arr[100], val;
    scanf("%d", &n);

    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    scanf("%d", &val);

    int k = 0;

    for (int i = 0; i < n; i++) {
        if (arr[i] != val) {
            arr[k] = arr[i];
            k++;
        }
    }

    printf("count = %d\n", k);

    for (int i = 0; i < k; i++)
        printf("%d ", arr[i]);

    return 0;
}
```

Student focus: Understand the loop and array indexing first. Optimization can be taught later.

13. LeetCode 283 — Move Zeroes

Simple idea: Solve the problem using basic arrays, loops and conditions.

```
#include <stdio.h>

int main() {
    int n, arr[100];
    scanf("%d", &n);

    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    int k = 0;

    for (int i = 0; i < n; i++) {
        if (arr[i] != 0) {
            arr[k] = arr[i];
            k++;
        }
    }

    while (k < n) {
        arr[k] = 0;
        k++;
    }

    for (int i = 0; i < n; i++)
        printf("%d ", arr[i]);

    return 0;
}
```

Student focus: Understand the loop and array indexing first. Optimization can be taught later.

14. LeetCode 268 — Missing Number

Simple idea: Solve the problem using basic arrays, loops and conditions.

```
#include <stdio.h>

int main() {
    int n, arr[100];
    scanf("%d", &n);

    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    int expected = n * (n + 1) / 2;
    int actual = 0;

    for (int i = 0; i < n; i++)
        actual += arr[i];

    printf("%d", expected - actual);

    return 0;
}
```

Student focus: Understand the loop and array indexing first. Optimization can be taught later.

15. LeetCode 136 — Single Number

Simple idea: Solve the problem using basic arrays, loops and conditions.

```
#include <stdio.h>

int main() {
    int n, arr[100];
    scanf("%d", &n);

    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    for (int i = 0; i < n; i++) {
        int count = 0;

        for (int j = 0; j < n; j++) {
            if (arr[i] == arr[j])
                count++;
        }

        if (count == 1) {
            printf("%d", arr[i]);
            return 0;
        }
    }

    return 0;
}
```

Student focus: Understand the loop and array indexing first. Optimization can be taught later.